



What should be added/subtracted to
a, b, c, d so that they are in
proportion?

Que: What should be added to each of 5, 9, 11, 18 to make them in proportion?



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Que: What should be added to each of 5, 9, 11, 18 to make them in proportion?

$$5+y : 9+y :: 11+y : 18+y$$

$$\frac{5+y}{9+y} = \frac{11+y}{18+y}$$

$$(5+y)(18+y) = (11+y)(9+y)$$

$$90 + 18y + 5y + y^2 = 99 + 9y + 11y + y^2$$

$$90 + 23y = 99 + 20y$$

$$23y - 20y = 99 - 90$$

$$3y = 9$$

$$y = 3$$

$$\frac{8}{3} = \frac{14}{21} = \frac{2}{3}$$

Que: What should be added to each of 19 , 37, 47, 86 to make them in proportion?



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Que: What should be added to each of 19, 37, 47, 86 to make them in proportion?

$$\frac{108}{21} \quad \text{5} \quad \checkmark$$

$$\begin{array}{r} 86 \\ 19 \\ \hline 105 \\ -84 \\ \hline 21 \end{array} \quad \begin{array}{r} 37 \\ 47 \\ \hline \end{array} \quad 84$$

$$19 \quad 37 \quad 47 \quad 86$$

$$19 \times 86 = 1634$$
$$37 \times 47 = 1739$$

$$\begin{array}{r} 1739 \\ -1634 \\ \hline 105 \end{array}$$

$$\frac{19}{37} \quad \frac{47}{86}$$

Que: What should be added to each of 19, 37, 47, 86 to make them in proportion?

$$\frac{19+24}{37+24} = \frac{47+91}{86+91}$$

$$\frac{4}{7} = \frac{4}{7} \checkmark$$

1-50

24

42

52

91

Que: What should be subtracted from each of a, b, c, d to make them in proportion?



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Que: What should be subtracted from each of a , b, c, d to make them in proportion?

$$\checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$
$$a-y, b-y, c-y, d-y$$

$$\frac{a-y}{b-y} = \frac{c-y}{d-y} \quad \checkmark$$

Que: What should be subtracted from each of 5, 9, 11, 18 to make them in proportion?



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Que: What should be subtracted from each of 5, 9, 11, 18 to make them in proportion?

$\frac{2}{6}$ $\frac{8}{15}$

$\frac{18}{12} = \frac{14}{8}$

$\frac{5-y}{9-y} = \frac{11-y}{18-y}$

$(18-y)(5-y) = (11-y)(9-y)$

$90 - 5y - 18y + y^2 = 99 - 9y - 11y + y^2$

$-23y + 20y = 99 - 90$

$-3y = 9$

$y = -3$

$5 - (-3) = 8$

$9 - (-3) = 12$

$11 - (-3) = 14$

$18 - (-3) = 21$